A First Course in Abstract Algebra

Section 46: Euclidean Domains

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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46 Euclidean Domains

Exercise 46.1: The square of absolute value

Yes. This map has nonnegative integer values. Ordinary integer division gives $a = bq + r$ with $0 \le r < |b|$, hence $r = 0$ or $r^2 < b^2$. For nonzero integers a, b, $a^2 \le a^2b^2$ because $b^2 \ge 1$. Both norm axioms hold.

Exercise 46.2: Degree over the integers

No. Divide $a = 1$ by $b = 2$ in $\mathbb{Z}[x]$. A nonzero remainder would have to satisfy $\deg r < \deg 2 = 0$, impossible. A zero remainder would say $1 = 2q$ in $\mathbb{Z}[x]$, also impossible. Degree division works over a field, not over arbitrary integer coefficients.

Exercise 46.3: Leading-coefficient size

No. Take dividend x and divisor 1. Every nonzero polynomial has norm at least 1, so a remainder with norm smaller than that of 1 must be zero; here zero remainder is possible, so this example alone is no obstruction. Instead divide x by x^2; zero remainder is impossible by degrees, while any nonzero remainder has leading-coefficient absolute value at least $1 = v(x^2)$. Thus the required division is impossible.

Exercise 46.4: A rational-valued proposal

No under the stated definition, since a Euclidean norm must take values in the nonnegative *integers*; $v(1/2) = 1/4$ does not. The monotonicity axiom also fails: $v(1) = 1 > v(1 \cdot 1/2) = 1/4$.

Exercise 46.5: A constant norm on a field

Yes. For $b \neq 0$ in $\mathbb{Q}$, set $q = a/b$ and $r = 0$. This gives the division axiom without needing a smaller norm. For nonzero a, b, $v(a) = 50 = v(ab)$ gives monotonicity.

Exercise 46.6: An explicit integer Bezout identity

Back substitution along the Euclidean algorithm gives

$$\begin{aligned} 22471 &= 6(3266) + 2875, & 3266 &= 2875 + 391, \\ 2875 &= 7(391) + 138, & 391 &= 2(138) + 115, \\ 138 &= 115 + 23, & 115 &= 5(23). \end{aligned}$$

Hence $23 = 138 - 115 = 3(138) - 391 = 3(2875) - 22(391) = 25(2875) - 22(3266) = 25(22471) - 172(3266)$. Direct multiplication verifies $561775 - 561752 = 23$, so $\lambda = 25, \mu = -172$.

Exercise 46.7: A larger integer gcd

Successive divisions give

$$\begin{aligned} 49349 &= 3(15555) + 2684, & 15555 &= 5(2684) + 2135,\\ 2684 &= 2135 + 549, & 2135 &= 3(549) + 488,\\ 549 &= 488 + 61, & 488 &= 8(61). \end{aligned}$$

The last nonzero remainder is 61, so a gcd is 61 (and -61 is an associate choice).

Exercise 46.8: Back substitution for the larger gcd

Using the remainders just found,

$$\begin{aligned} 61 &= 549 - 488 = 4(549) - 2135\\ &= 4(2684) - 5(2135) = 29(2684) - 5(15555)\\ &= 29(49349) - 92(15555). \end{aligned}$$

Thus $\lambda = 29, \mu = -92$. Checking the final equality gives $1431121 - 1431060 = 61$.

Exercise 46.9: A polynomial gcd

Let f, g denote the displayed degree-ten and degree-six polynomials. Exact polynomial division gives

$$\begin{aligned} f &= (x^4 - 2x)g + r,\\ r &= -x^4 - 3x^3 - 2x^2 - 5x + 3\\ &= (-x - 3)(x^3 + 2x - 1),\\ g &= (x^3 - 3x^2 + x - 2)(x^3 + 2x - 1). \end{aligned}$$

The cofactors $-x - 3$ and $x^3 - 3x^2 + x - 2$ are relatively prime: substituting $x = -3$ into the latter gives $-59 \neq 0$, so its linear possible common factor does not divide it. Consequently a monic gcd is $x^3 + 2x - 1$. The first division and the displayed factorizations can be checked by multiplying out; they also show directly why no further factor is common.

Exercise 46.10: A gcd of many elements

Compute $d_2 = \gcd(a_1, a_2)$, then $d_3 = \gcd(d_2, a_3)$, and continue until d_n. At each step, an element divides d_k exactly when it divides all $a_1, \ldots, a_k$. This follows by induction from the defining common-divisor property of a gcd. Therefore d_n is a gcd of the entire list. Ignore zero entries, or use $\gcd(a, 0) = a$ up to associates.

Exercise 46.11: Four integers by successive gcds

First $2178 = 5(396) + 198$ and $396 = 2(198)$, so the first gcd is 198. Since $792 = 4(198)$, it remains 198 after the third entry. Finally $726 = 3(198) + 132$, $198 = 132 + 66$, and $132 = 2(66)$. Thus the positive gcd is 66.

Exercise 46.12: The ideal of even constant terms

(a) $\mathbb{Z}[x]$ is a UFD because $\mathbb{Z}$ is a UFD and Gauss's theorem preserves this property on adjoining x.

(b) Let I consist of polynomials with even constant term. Differences preserve this property; multiplying such a polynomial by any integer polynomial multiplies its constant term by an integer, so it remains even. Thus I is an ideal, equal to $(2, x)$.

(c) If $I = (h)$, then h divides both 2 and x. Dividing 2 forces h to be a constant integer divisor of 2; dividing x then forces $h = \pm 1$. This would put 1 in I, contrary to its odd constant term. Hence I is not principal.

(d) A Euclidean domain is a PID. Since this ring is not a PID, it is not Euclidean.

Exercise 46.13: Ten statements about Euclidean domains

(a) True: use a nonzero element of smallest norm in an ideal and divide by it.

(b) False: for example $\mathbb{Z}[(1 + \sqrt{-19})/2]$ is a PID but is not Euclidean. The implication has no converse in general.

(c) True, since every PID is a UFD.

(d) False: $\mathbb{Z}[x]$ is a UFD but not a PID, hence not Euclidean.

(e) True: in the field $\mathbb{Q}$, $1/2$ divides every rational, and every common divisor divides it.

(f) True: successive nonnegative integer norms strictly decrease until the remainder is zero.

(g) True: apply $v(a) \leq v(ab)$ with $a = 1$.

(h) False: in $\mathbb{Z}$, $v(-1) = v(1)$ for $v(n) = |n|$, although $-1 \neq 1$.

(i) True: equality $v(a) = v(1)$ would make a a unit by the norm characterization, proved in Exercise 46.16.

(j) True, with the degree norm and polynomial division over F.

Exercise 46.14: A norm measures, rather than changes, arithmetic

No. Addition, multiplication, units, divisibility, ideals, and gcds are defined by the domain operations. A chosen norm helps find quotients and remainders, but does not alter these operations. Different permissible norms can change which division choices the algorithm makes; the final gcds still differ only by units.

Exercise 46.15: Equal norms for associates

If $a = bu$ with u a unit, monotonicity gives $v(b) \leq v(bu) = v(a)$. Also $b = au^{-1}$, so $v(a) \leq v(b)$. Thus $v(a) = v(b)$. The norms are defined here for nonzero associates.

Exercise 46.16: Strict growth detects a nonunit factor

If b is a unit, a and ab are associates, so their norms are equal. Conversely, suppose $v(a) = v(ab)$. Divide a by ab: write $a = abq + r$, with $r = 0$ or $v(r) < v(ab) = v(a)$. But $r = a(1 - bq)$, so if $r \neq 0$, monotonicity gives $v(a) \leq v(r)$, a contradiction. Thus $r = 0$, and cancellation of $a \neq 0$ yields $1 = bq$. Therefore b is a unit. As $v(a) \leq v(ab)$ always, nonunit b is exactly the strict-inequality case.

Exercise 46.17: The nonunits need not form an ideal

The proposed set is the nonunits together with zero, by the unit characterization. In $\mathbb{Z}$ with absolute-value norm, both 2 and 3 lie in it, but $3 - 2 = 1$ does not. It is therefore not an additive subgroup and is not an ideal in general.

Exercise 46.18: Every field is Euclidean

Define $v(a) = 0$ for every nonzero field element. Divide a by nonzero b by taking $q = ab^{-1}$ and $r = 0$. This satisfies the division axiom, and $v(a) = v(ab) = 0$ satisfies monotonicity. Thus every field is Euclidean.

Exercise 46.19: Changing the scale of a norm

(a) Since $v(a) \geq v(1)$, the assumption $s + v(1) \geq 0$ ensures $\eta(a) \geq 0$. Adding the same integer to two norms preserves both their strict and weak inequalities, so it preserves the two axioms.

(b) Multiplication by a positive integer t preserves nonnegative integer values and all inequalities, so $\lambda(a) = tv(a)$ is a Euclidean norm.

(c) Set $\mu(a) = 1 + 100(v(a) - v(1))$. This is a positive integer norm by (a),(b); it equals 1 on units. A nonunit has integer norm strictly larger than $v(1)$, so $\mu(a) \geq 101 > 100$.

Exercise 46.20: Least common multiples

Let $I = (a) \cap (b)$, the ideal of common multiples. It is nonzero because $ab \in I$ and $ab \neq 0$. A Euclidean domain is a PID, so $I = (c)$ for some nonzero c. Membership of c in both ideals says $a \mid c, b \mid c$. Every common multiple belongs to (c), hence is divisible by c. These are exactly the required lcm properties. Multiplying c by a unit gives another lcm.

Exercise 46.21: Generating the additive integers

The subgroup generated by r, s is $\{ur + vs : u, v \in \mathbb{Z}\}$. Bezout's theorem identifies this with $d\mathbb{Z}$, where $d = \gcd(r, s) > 0$: every combination is divisible by d, and a Bezout combination equals d. It is all $\mathbb{Z}$ precisely when $d = 1$.

Exercise 46.22: A solvable congruence when the coefficient is coprime

If $\gcd(a, n) = 1$, choose integers u, v with $ua + vn = 1$. Multiplying by b gives $a(ub) + n(vb) = b$. Thus $x = ub$ satisfies $ax \equiv b \pmod{n}$. In $\mathbb{Z}_n$ this says the nonzero residue of a is invertible when it is coprime to n.

Exercise 46.23: The general congruence criterion

Put $d = \gcd(a, n) > 0$. If $ax \equiv b \pmod{n}$, then $b = ax - kn$ for some integer k, and $d \mid b$. Conversely, if $b = dc$, multiply a Bezout identity $ua + vn = d$ by c. It gives $a(uc) \equiv b \pmod{n}$. In the residue ring $\mathbb{Z}_n$, the principal ideal generated by $\bar{a}$ consists exactly of residues represented by multiples of d; solvability asks whether $\bar{b}$ lies in this ideal.

Exercise 46.24: Constructing a solution

Run the extended Euclidean algorithm to obtain $d = ua + vn$. If $d \nmid b$, no solution exists. Otherwise $x_0 = u(b/d)$ is a solution; all solutions are $x = x_0 + k(n/d)$. For $22x \equiv 18 \pmod{42}$, $d = 2$ and $2 = 2(22) - 42$. Thus $x_0 = 2(18/2) = 18$. Check: $22(18) - 18 = 378 = 9(42)$. All integer solutions are $x = 18 + 21k$; modulo 42 these are 18 and 39.